

Microsoft IQ Deep Dive with Python

July 28: Foundry IQ

July 29: Work IQ

July 30: Fabric IQ



Stay
GROUNDED

Register at aka.ms/IQDeepDivePython/series



Microsoft IQ Deep Dive: Foundry IQ

aka.ms/iqdeepdive/slides/foundryiq



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Today we'll cover...

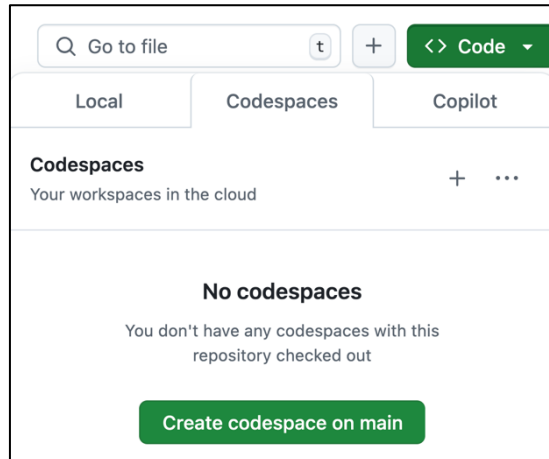
- Foundry IQ knowledge bases
- Data ingestion process
- Web IQ for web retrieval
- Foundry IQ inside agents
- Foundry IQ in Foundry Toolbox
- Deploying agents with Foundry IQ

Want the code?

1. Open this GitHub repository:

<https://aka.ms/iqdeepdive>

2. Use "Code" button to create a GitHub Codespace:



3. Wait a few minutes for Codespace to start up 🕒



All code samples require deployment and an Azure account.

Microsoft IQ



Microsoft IQ Platform

Unified intelligence for enterprise AI



Work IQ

How your employees work



Fabric IQ

How your business operates



Web IQ

How you connect to web intelligence



Foundry IQ

How your agents unlock knowledge

Context on people,
collaboration, and workflows

**Deep dive:
Session 2**

Context on business
entities, systems of
record, and actions

**Deep dive:
Session 3**

Context from the web,
news, images and video

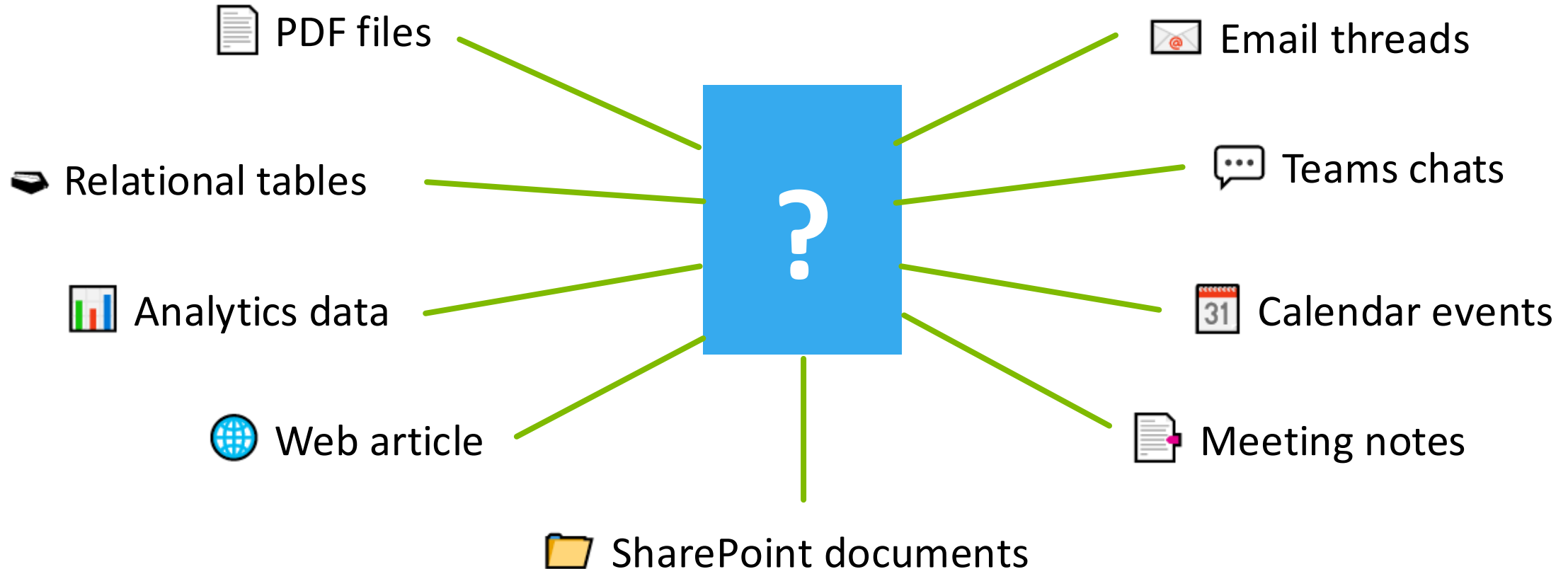
**Covering today:
Session 1**

Context on policies,
authoritative documents, and
knowledge bases

Foundry IQ

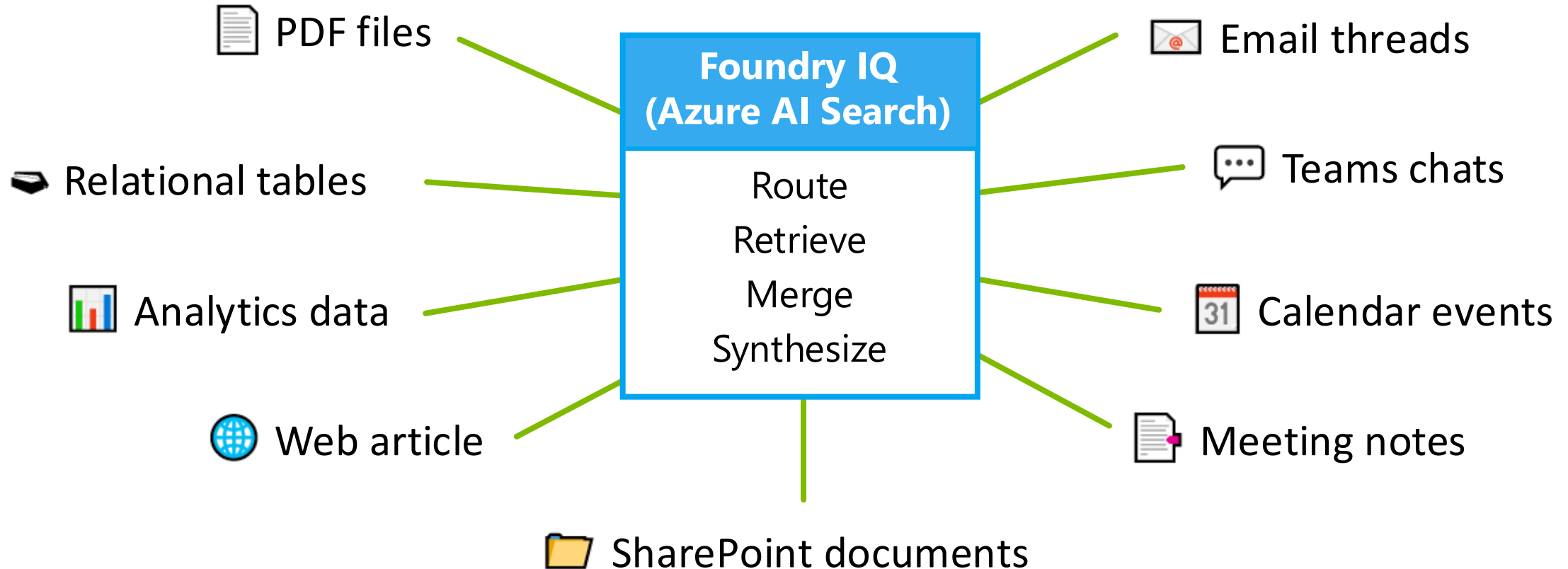


Knowledge is everywhere

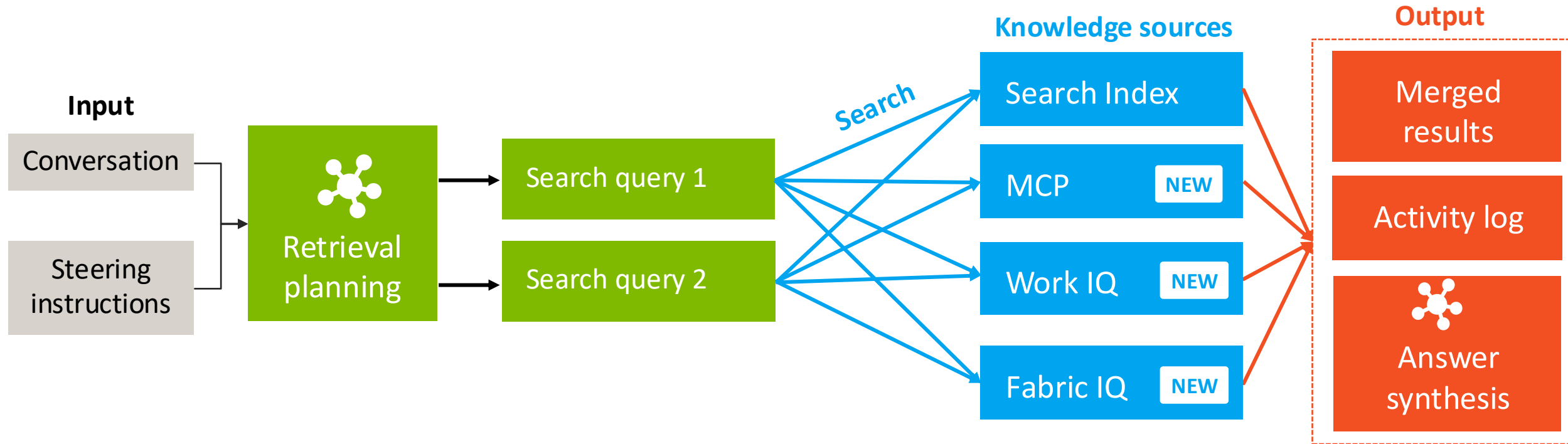


...But how does an agent get to it?

Foundry IQ makes knowledge agent-ready



Inside a Foundry IQ knowledge base



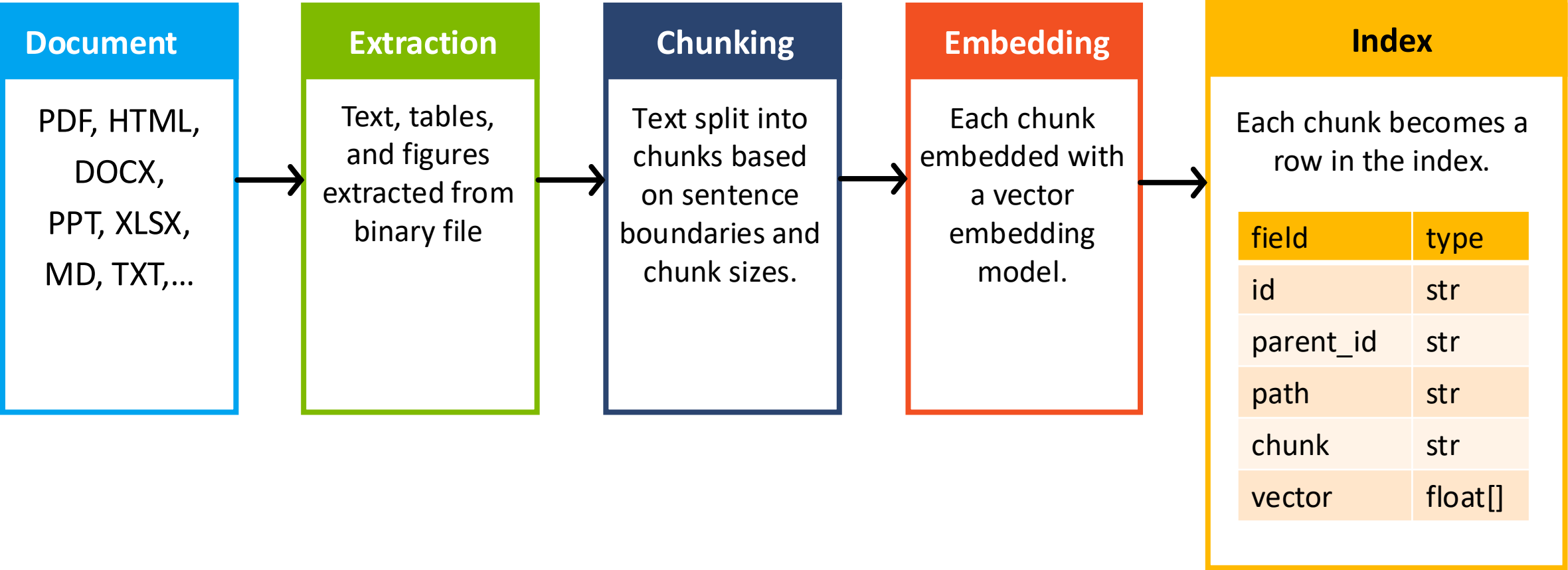
One question. Every source. One answer.

Indexed knowledge



Data ingestion for unstructured data

For unstructured documents, we need a data ingestion flow:



Two ways to add indexed knowledge

Choose based on how much control you need over ingestion:

File Knowledge Source

You bring: **Files**



Automatic ingestion pipeline:

Extract → Chunk → Embed → Index

- Zero ingestion code
- Just upload the files and go
- Fixed index schema
- Less flexibility

Best for: quick start, prototyping,
small documents

Indexed Knowledge Source

You bring: **A data source (Blob, DB, ...)**



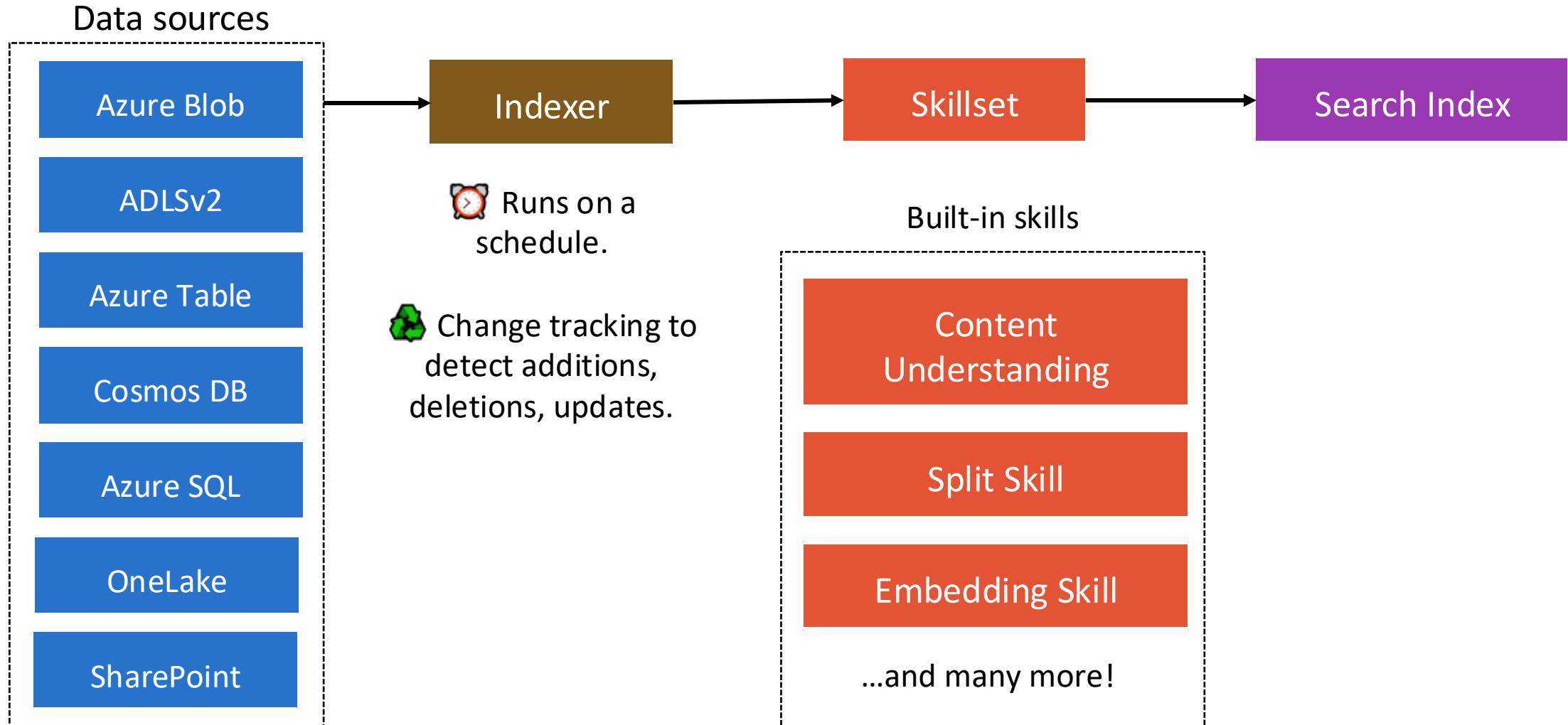
Your custom configured ingestion pipeline:

Data source → Indexer → Skillset → Index

- Full schema and pipeline control
- Custom enrichments and skillsets
- Works with existing data sources
- More custom code to maintain

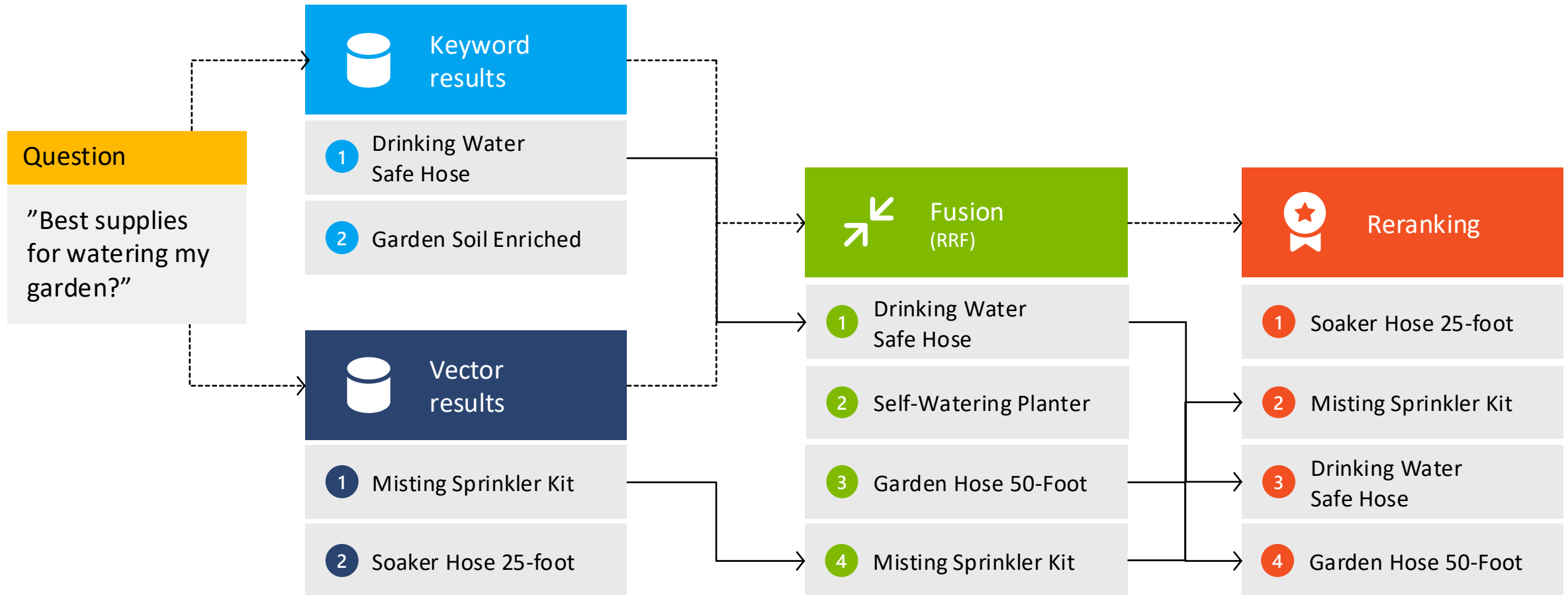
Best for: production data, custom schemas,
domain-specific customizations

Indexed Knowledge Ingestion



Hybrid retrieval on indexed knowledge

A complete search stack gives the best results. Hybrid > pure vector or keyword!



Knowledge bases



Knowledge base retrieval reasoning effort

MINIMAL

LOW

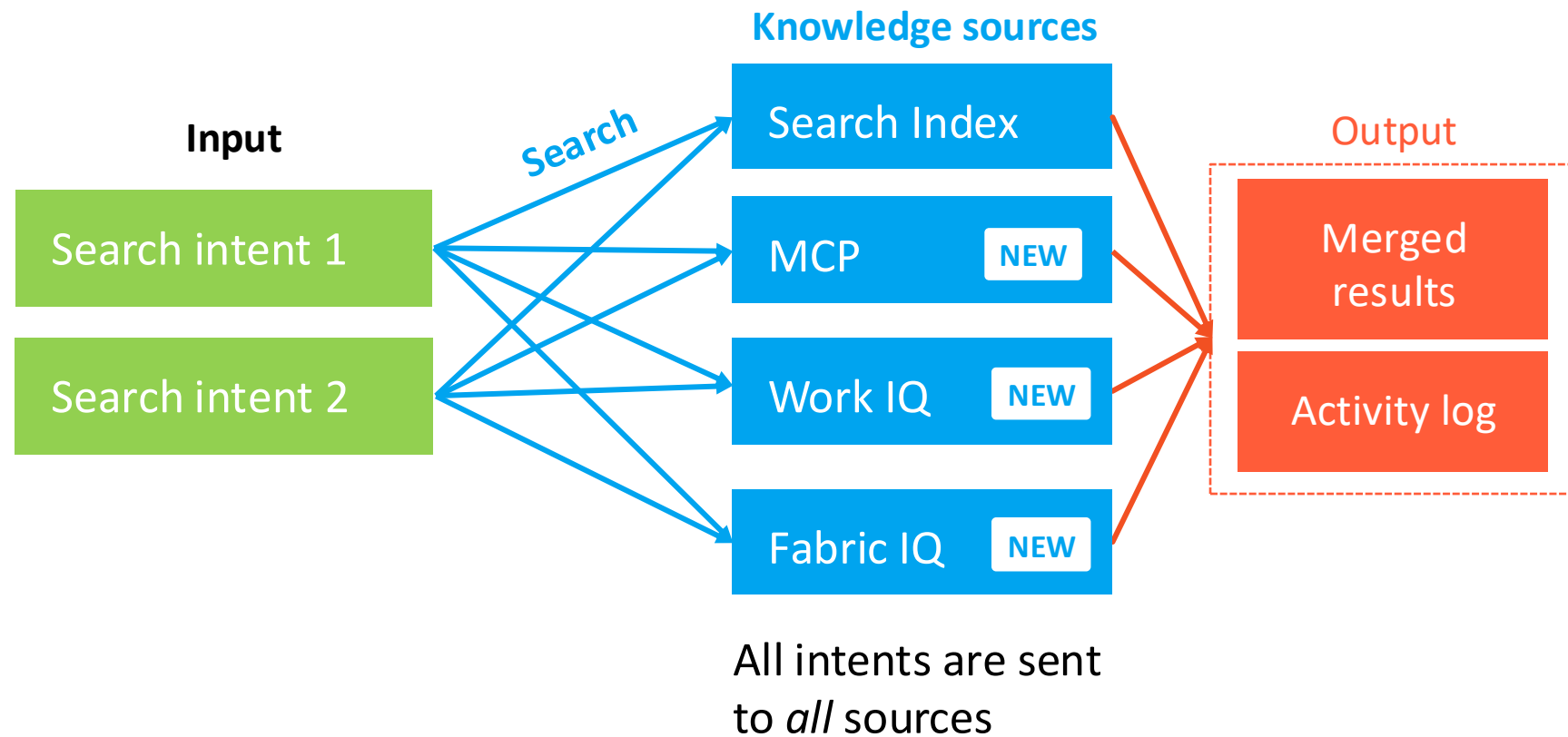
MEDIUM

Low latency
Less model usage
Lowest cost

Highest quality
Increased stages
More agentic

Knowledge base with minimal effort

Minimal effort does not use any LLM, but still uses re-ranker model for merging.



Example of minimal effort

Input

"what's best Zava paint for bathroom walls?"

* You may optionally pass in multiple intents.

Output

Activity log

Step	Details
Search ""what's best Zava paint for bathroom walls?"	Source: Search index Results: 8
Search ""what's best Zava paint for bathroom walls?"	Source: SharePoint Results: 2

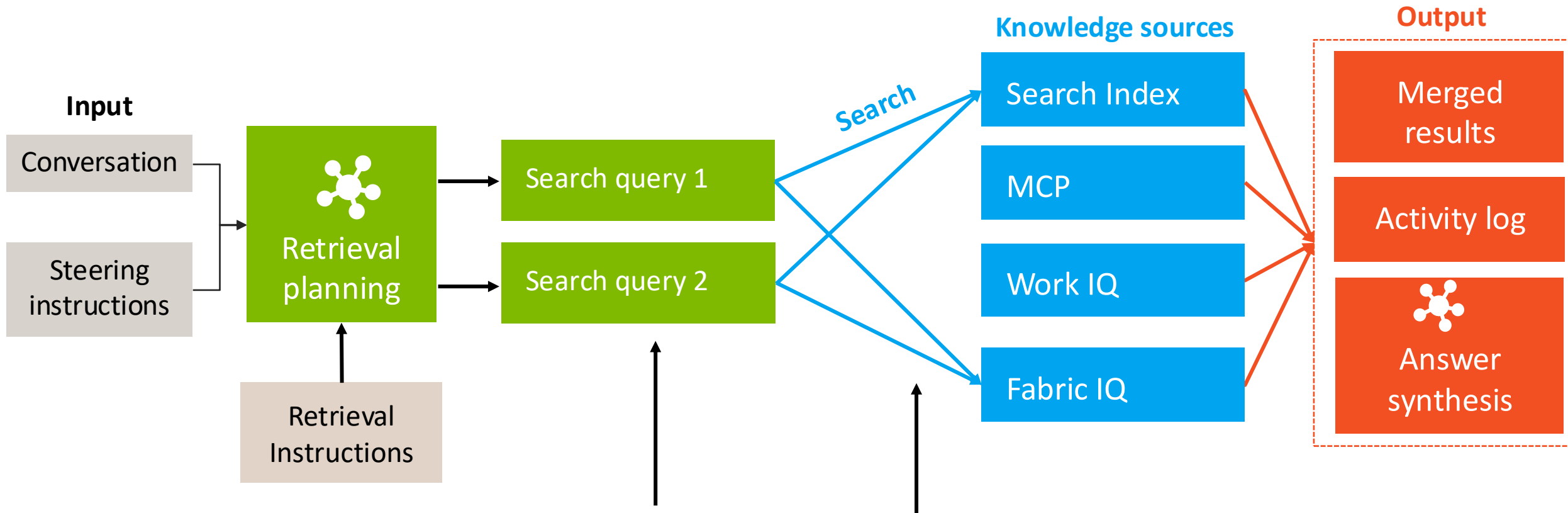
Intent is sent to all sources.

Merged results

.pdf
Interior Semi-Gloss Paint, price \$47.0\r\n\r\n ## Brand\r\nZavaTech Hardware...
🏆 Re-ranker score: 2.95
ZavaBathroomPost.pdf
Refresh Your Bathroom with Confidence: Why Semi-Gloss Paint Makes All the Difference When it comes to painting projects, bathrooms are...
🏆 Re-ranker score: 3.07

Knowledge base with low effort

Low effort uses an LLM for retrieval planning and optional answer synthesis.



Retrieval planning generates queries *and* selects sources.

Example of low effort with knowledge source selection

Input

What Zava paint can I use to paint my bathroom and how much does it cost?

Output

Activity log

Step	Details
Query planning	Input tokens: 2113 Output tokens: 75
Search "Zava paint suitable for bathroom use"	Source: Search index Results: 6
Search "Zava paint prices"	Source: Search index Results: 14
Agentic reasoning	Reasoning: 20K Effort: Low
Model answer synthesis	Input tokens: 7921 Output tokens: 108

Merged results

PFIP0000003.pdf

Interior Semi-Gloss Paint, price \$47.0\r\n\r\n ##
Brand\r\nZavaTech Hardware...

🏆 Re-ranker score: 3.03

PFIP0000005.pdf

One-Coat Interior Paint, price \$50.00\r\n\r\n ##
Brand\r\nZavaTech Hardware

🏆 Re-ranker score: 2.95

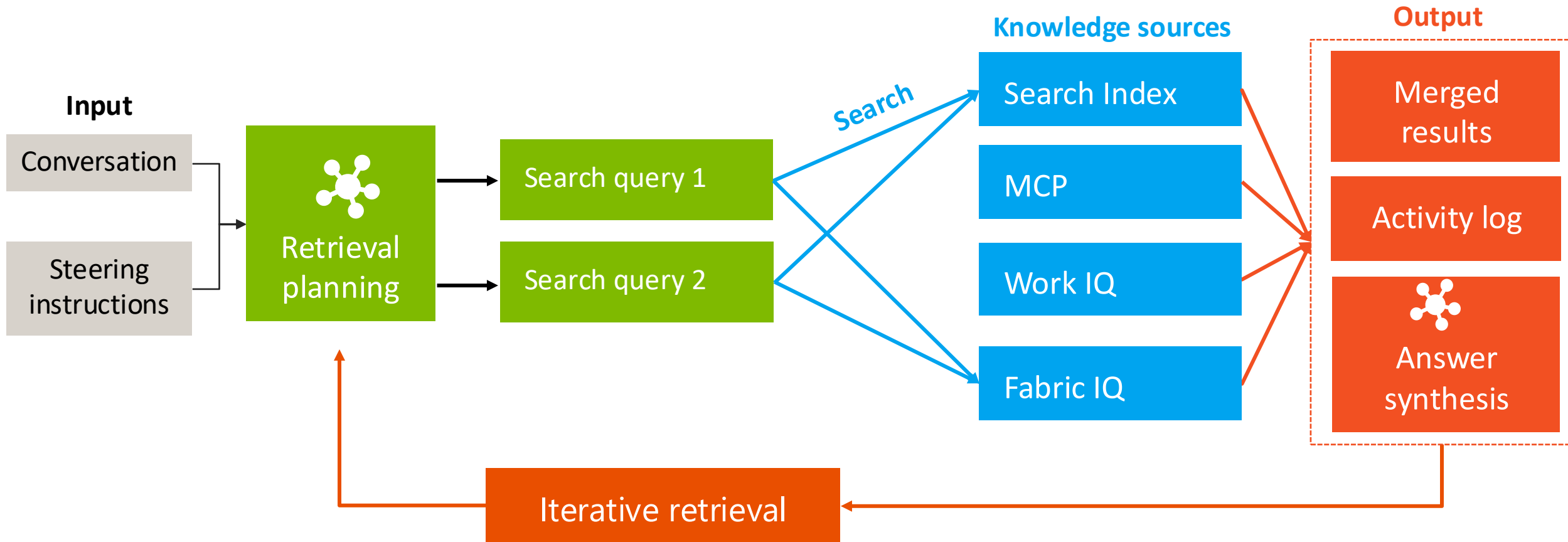
 Answer synthesis

A semi-gloss interior paint is suitable for use in a bathroom because it offers moisture resistance, is washable, and stands up well to humidity and frequent cleaning. This type of paint is specifically recommended for bathrooms, kitchens, and trim work due to its durability and ease of maintenance [ref_id:1]. The typical cost for a can of Interior Semi-Gloss Paint from ZavaTech Hardware Solutions is \$47.00 [ref_id:2].

Answer synthesis is *optional*.

Knowledge base with medium effort

Medium effort adds a second retrieval, only when it still needs more context to answer the query.



Medium effort adds a second retrieval, only when it still needs more context to answer the query.

Example of medium effort

Input

Explain how to paint my house most efficiently. Then give me a list of the Zava products and prices for each supply

Output

Activity log

Step	Details
Query planning	Input tokens: 1484 Output tokens: 115
Search "efficient house painting"	Source: manuals
Search "Zava paint supplies prices"	Source: manuals
Search "efficient house painting"	Source: web
Search "Zava paint prices"	Source: web
Query planning	Input tokens: 1169 Output tokens: 249
Search "paint brushes rollers trays"	Source: manuals
Search "paint brushes rollers trays"	Source: web
Agentic reasoning	Reasoning: 65K Effort: Medium
Model answer synthesis	Input tokens: 12676 Output tokens: 196

Merged results

PFEP000007.pdf

Exterior Acrylic Paint, price \$57.0\r\n\r\n## Brand\r\nZavaTech...

🏆 Re-ranker score: 2.85

francoisestmoi.com

Url: http://francoisestmoi.com/diy/top-5-ways-to-paint-more-efficiently

Title: Top 5 Ways to Paint More Efficiently

qualitypreferred.com

Url: https://www.qualitypreferred.com/choos...ng-the-right-method-the-best-way-to-paint-a-house

Title: The Best Way to Paint a House

 Answer synthesis

The most efficient way to paint a house is to work from the top down: start by prepping all surfaces, then paint ceilings first, followed by trim and baseboards, and finish with the walls, using rollers for large areas and brushes or pads for detail work to minimize drips and touch-ups [ref_id:5][ref_id:3][ref_id:4]. For exterior painting, clean and repair surfaces, prime exposed areas, and always paint from high to low, finishing with doors and trim [ref_id:5][ref_id:6].

Below is a list of Zava paint supplies:
* Exterior Acrylic Paint: \$57.0 [ref_id:1]

Foundry IQ multi-source knowledge base

Create a knowledge base with multiple knowledge sources:

```
knowledge_base = KnowledgeBase(  
    name="multisource-search-knowledge-base",  
    description="Multi-source knowledge base over HR and health document indexes",  
    models=[KnowledgeBaseAzureOpenAIModel(azure_open_ai_parameters=aoai_params)],  
    knowledge_sources=[KnowledgeSourceReference(name="hrdocs-knowledge-source"),  
                       KnowledgeSourceReference(name="healthdocs-knowledge-source")],  
    retrieval_reasoning_effort=KnowledgeRetrievalLowReasoningEffort(),  
    output_mode=KnowledgeRetrievalOutputMode.ANSWER_SYNTHESIS  
)  
index_client.create_or_update_knowledge_base(knowledge_base)
```

Query the knowledge base with user query:

```
retrieval_request = KnowledgeBaseRetrievalRequest(  
    messages=[KnowledgeBaseMessage(role="user",  
                                    content=[KnowledgeBaseMessageTextContent(text="How many vacation weeks do we get?")])],  
    include_activity=True)  
  
result = knowledge_base_client.retrieve(retrieval_request=retrieval_request)
```



Full example: [notebooks/foundryiq-basic.ipynb](https://github.com/FoundryIQ/notebooks/foundryiq-basic.ipynb)

Web IQ



Microsoft Web IQ

A web search service designed specifically for agents:

Web Search

News

Finance

Video

Sports

Browse

Very low latency

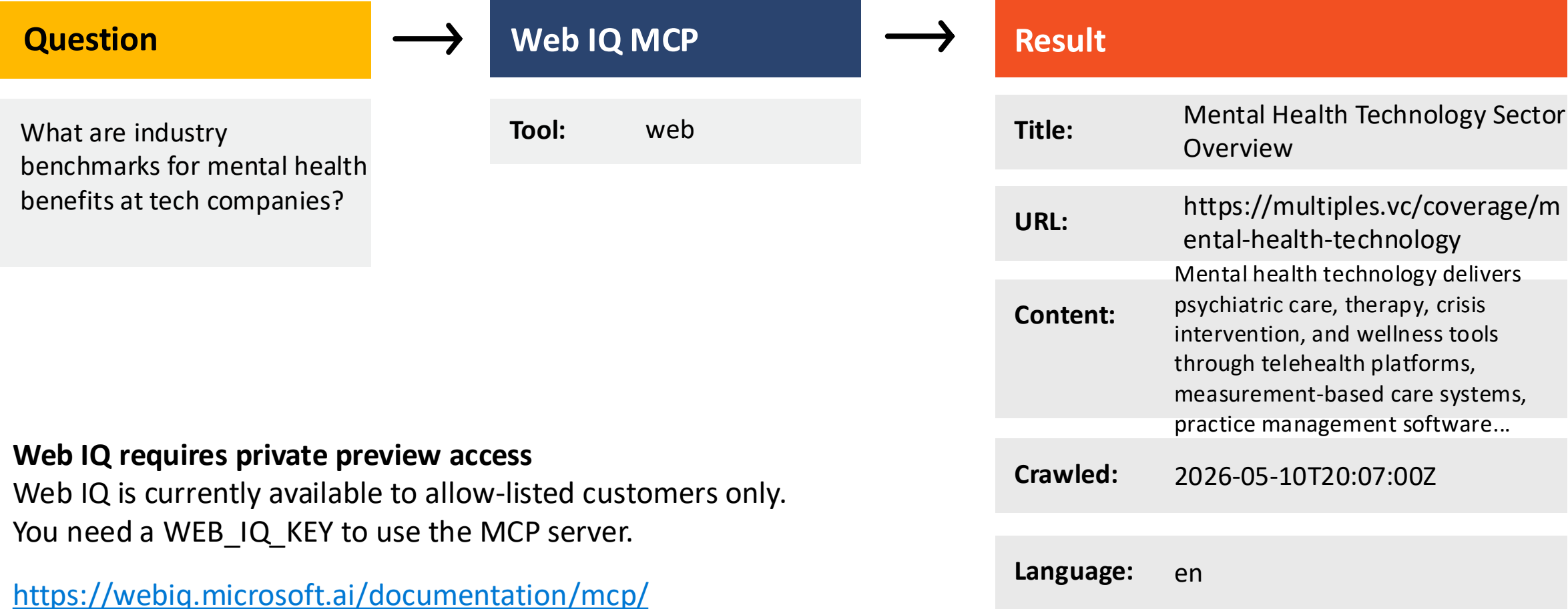
~164ms P95
2.5x faster than today's best alternative

Full payload

Results includes full Markdown, HTML, or summaries, all in one response.

<https://webiq.microsoft.ai/>

Retrieval with Microsoft Web IQ MCP server



Foundry IQ + Web IQ knowledge source

Create an MCP knowledge source for WebIQ MCP server:

```
web_knowledge_source = McpServerKnowledgeSource(  
    name="web-knowledge-source", description="Web IQ (live web search)",  
    mcp_server_parameters=McpServerKnowledgeSourceParameters(  
        server_url="https://api.microsoft.ai/v3/mcp",  
        authentication=McpServerStoredHeadersAuthentication(  
            stored_headers_parameters=McpServerStoredHeadersParameters({"headers": {"x-apikey": WEB_IQ_KEY}})),  
        tools=[McpServerTool(name="web", output_parsing=McpServerAutoOutputParsing())])
```

Create a knowledge base that includes the Web IQ MCP knowledge source:

```
knowledge_base = KnowledgeBase(  
    name="multisource-web-knowledge-base",  
    description="Multi-source knowledge base combining indexed company documents and live web results",  
    models=[KnowledgeBaseAzureOpenAIModel(azure_open_ai_parameters=aoai_params)],  
    knowledge_sources=[KnowledgeSourceReference(name="web-knowledge-source"),  
        KnowledgeSourceReference(name="hrdocs-knowledge-source")])
```

Query the knowledge base with user query:

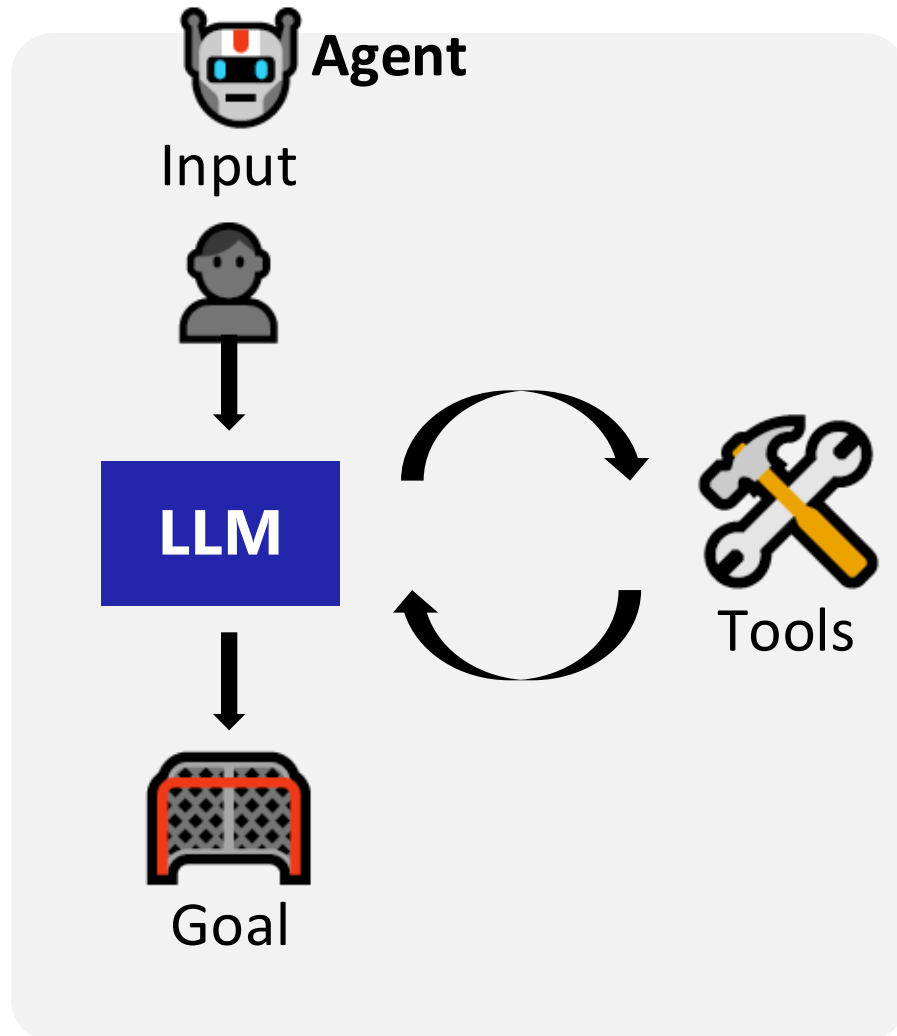
```
result = knowledge_base_client.retrieve(KnowledgeBaseRetrievalRequest(  
    messages=[KnowledgeBaseMessage(role="user", content=[KnowledgeBaseMessageTextContent(text=question)])])
```

 Full example: [notebooks/foundryiq-webiq.ipynb](https://github.com/microsoft/foundryiq-webiq.ipynb)

Foundry IQ inside Agents



What is an agent?



An **AI agent** uses an **LLM** to run **tools** in a loop to achieve a **goal**.



You're an internal HR helper assistant.



How long til benefits sign-up closes?

LLM

`get_current_datetime()`



Tues, July 28, 2026, 10:15 AM

LLM

`search_kb("benefits deadline")`

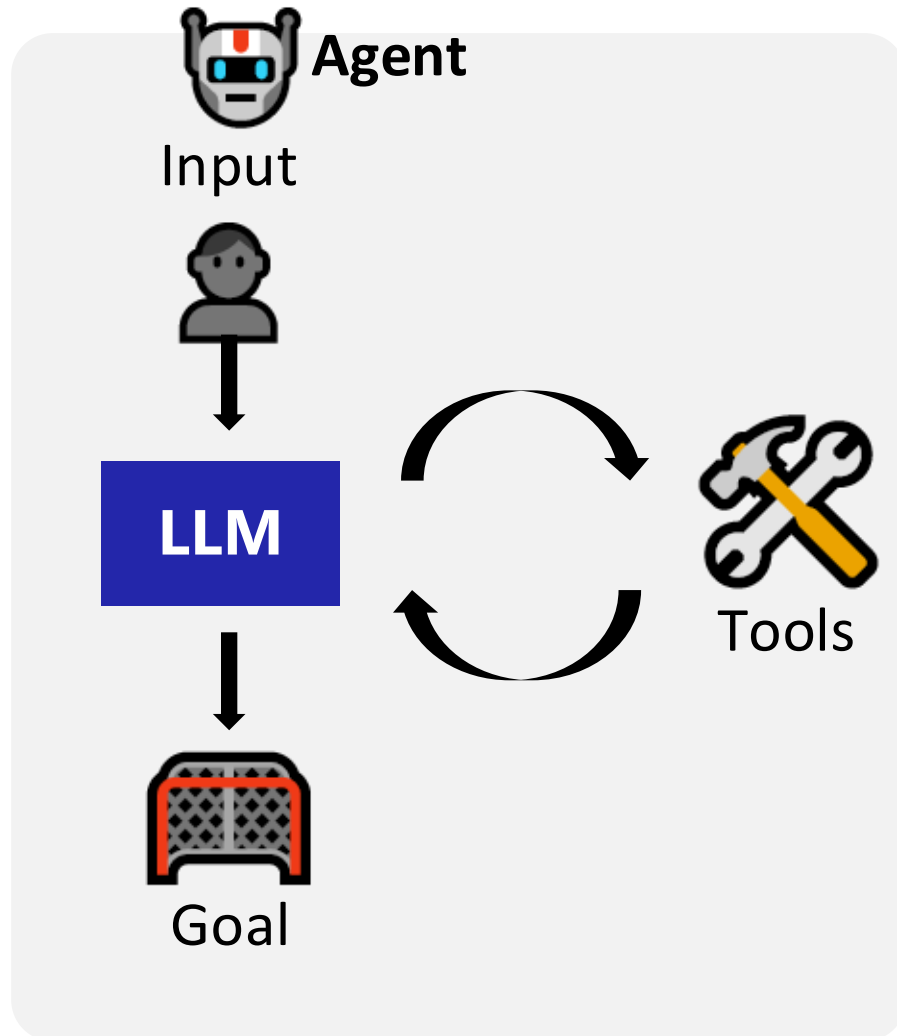


Enrollments deadline in August 1st...

LLM

You have only 3 days to sign up!

Using Foundry IQ as a tool for AI agents



Option 1:

Write a custom tool that calls the KB via API

👍 Full control over parameters and response

👤 Less portability across agents

Option 2:

Point agent directly at the KB MCP endpoint

👍 Less code to maintain

👤 Can't customize retrieval parameters

👤 Can't access references and activity log

Option 3:

Point agent at Foundry Toolbox with KB MCP

👍 Easier to add on more tools

👍 Managed authorization flow for auth'd sources

👤 Same drawbacks as MCP server

Which retrieval effort to use with agents?

The deployed agents in this repository use a **minimal** knowledge base, then let the Python agent itself handle query rewriting and iteration.



Using Foundry IQ as a tool: Custom code

Define a tool that uses KB SDK:

```
@tool
async def retrieve_company_knowledge(queries: list[str]) -> dict[str, Any]:
    """Retrieve grounded company and HR information from the Foundry IQ knowledge base."""
    request = KnowledgeBaseRetrievalRequest(
        intents=[KnowledgeRetrievalSemanticIntent(search=query) for query in queries],
        include_activity=True)
    result = await knowledge_base_client.retrieve(request)
    return {"response": serialize_models(result.response),
            "references": serialize_models(result.references),
            "activity": serialize_models(result.activity)}
```

Create an agent with the tool:

```
agent = Agent(client=client, name="InternalHRApiHelper",
              instructions="You are an internal HR helper.",
              tools=[retrieve_company_knowledge])
```



Full example: <src/agent-foundryiq-api/main.py>

Using Foundry IQ as a tool

DEMO

What benefits are available, and when do I need to enroll?

retrieve_company_knowledge

Input

```
{"queries":["What employee benefits are available? Provide benefits"]}
```

Output

```
{"response": [{"content": [{"type": "text", "text": "[{"ref_id": 0, "content": "Northwind Health. This coverage provides financial protection to employees in the event of a work-related injury or illness, including medical treatment and lost wages. In the event of a work-related injury or illness, employees are eligible to receive benefits such as: Medical care: This includes doctor visits, hospital care, and other treatment deemed necessary by a physician. Wage replacement: This includes a portion of wages lost due to the injury or illness. Vocational rehabilitation: This includes education, retraining, and job placement assistance. Death benefits: This includes a lump sum payment to the surviving spouse or dependents in the event of a work-related death. In order to receive these benefits, employees must report the injury or illness to their supervisor as soon as possible."}]}]}
```

> get_enrollment_deadline_info

> get_current_date

Here's a grounded summary of the benefits I could confirm and the enrollment timing:

Available benefits

- **Health insurance plans**
 - **Northwind Health Plus:** comprehensive medical, vision, and dental coverage, including preventive/wellness services, prescription drugs.

Run local agent:

```
azd ai agent run agent-foundryiq-api
```

Ask question with playground or CLI:

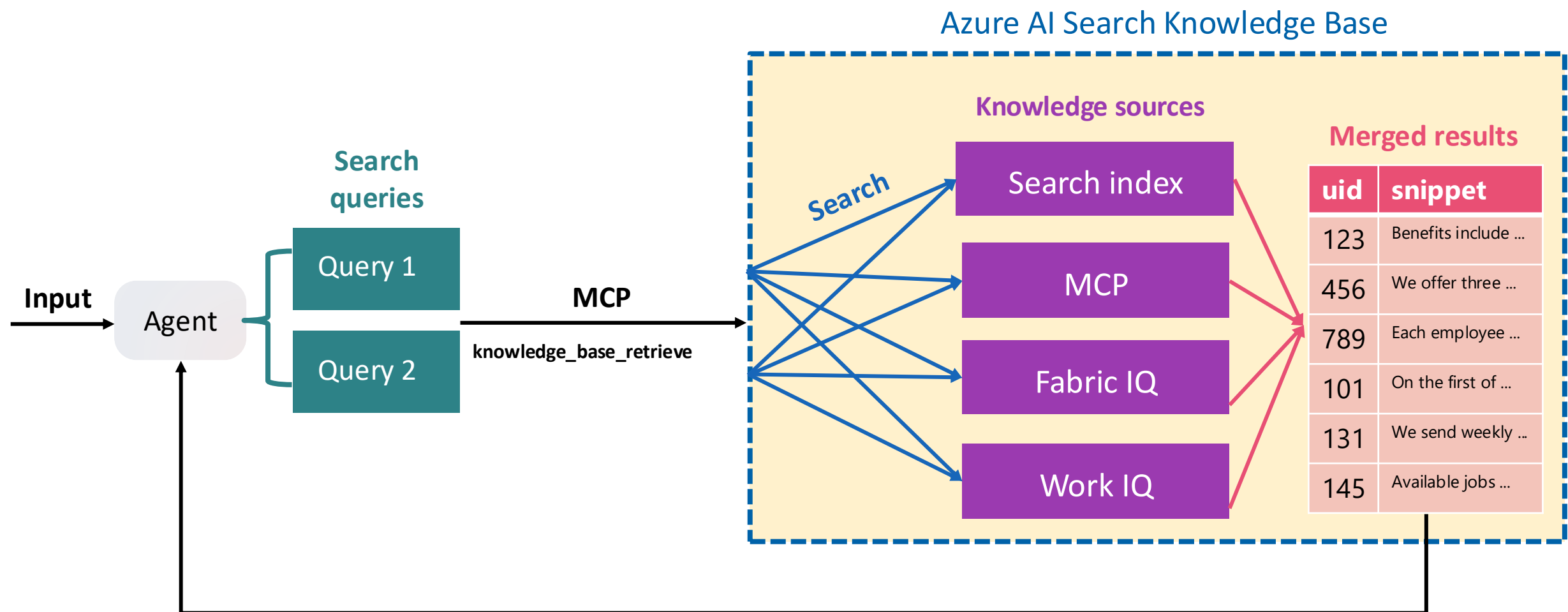
```
azd ai agent invoke agent-foundryiq-api --local "What benefits are available, and when do I need to enroll?"
```



Full example: [src/agent-foundryiq-api/main.py](https://github.com/microsoft/agent-foundryiq-api/blob/main/main.py)

Using Foundry IQ as a tool: MCP server

The MCP endpoint returns merged results with citations (but no activity log or separate references).



Using Foundry IQ as a tool: MCP server

```
knowledge_base_endpoint = (f"{SEARCH_ENDPOINT}/knowledgebases/{KNOWLEDGE_BASE_NAME}  
                           /mcp?api-version=2026-05-01-preview")  
  
knowledge_base_http_client = httpx.AsyncClient(  
    auth=AzureTokenCredentialAuth(credential, SEARCH_SCOPE))  
  
knowledge_base_mcp_tool = MCPStreamableHTTPTool(  
    name="knowledge-base",  
    url=knowledge_base_endpoint,  
    http_client=knowledge_base_http_client,  
    allowed_tools=["knowledge_base_retrieve"],  
    load_prompts=False)  
  
agent = Agent(  
    client=client,  
    name="InternalHRHelper",  
    instructions="You are an internal HR helper.",  
    tools=[knowledge_base_mcp_tool])
```



Full example: [src/agent-foundryiq-mcp/main.py](https://github.com/agent-foundryiq-mcp/main.py)

Using Foundry IQ as an MCP server

DEMO

What benefits are available, and when do I need to enroll?

knowledge_base_retrieve

Input

```
{"queries":["What benefits are available, and when do I need to enroll?"]}
```

Output

```
[{"ref_id":0,"content":"# Contoso\n\nDisclaimer: This content is generated by AI and may not accurately represent factual information about any real entity. Use this information with caution and verify details from reliable sources.\n\n## History\n\nContoso, a pioneering force in the industry, was founded in 1985 by visionary entrepreneurs with a passion for innovation. Over the years, the company has played a pivotal role in shaping the landscape of consumer electronics.\n\nYear | Milestone\n-----\n\n1985 | Company founded with a focus on cutting-edge technology\n1990 | Launched the first-ever handheld personal computer\n2000 | Introduced groundbreaking advancements in AI and robotics\n2015 | Expansion into sustainable and eco-friendly product lines\n\n## Company Overview\n\nAt Contoso, we take pride in fostering a dynamic and inclusive workplace. Our dedicated team of experts collaborates to create innovative
```

> get_enrollment_deadline_info

Here's what I could confirm from the available benefits information:

Available benefits

- **Health insurance through Northwind**
 - **Northwind Standard:** medical, vision, and dental coverage; preventive care; prescription drugs.
 - **Northwind Health Plus:** medical, vision, and dental coverage; prescription drugs; mental health and substance abuse coverage;

Run local agent:

```
azd ai agent run agent-foundryiq-mcp
```

Ask question with playground or CLI:

```
azd ai agent invoke agent-foundryiq-mcp --local "What benefits are available, and when do I need to enroll?"
```

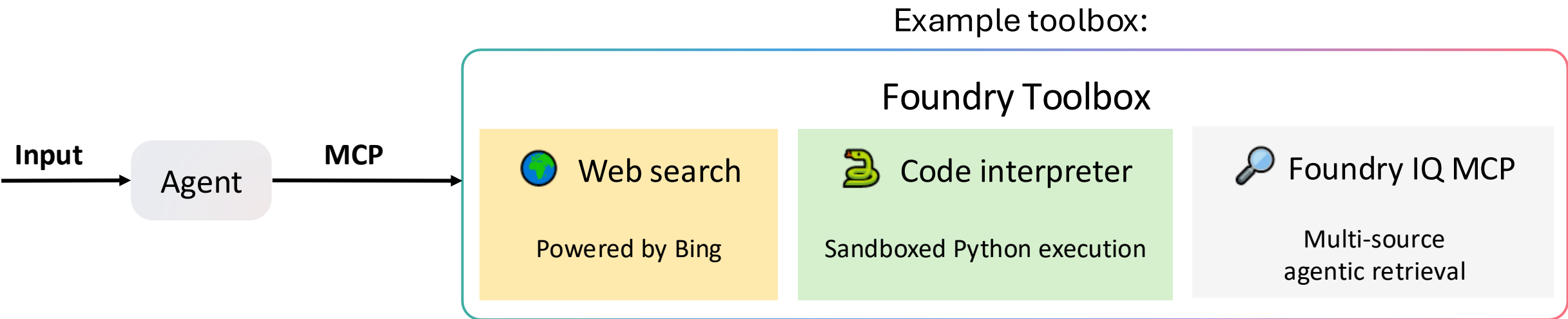
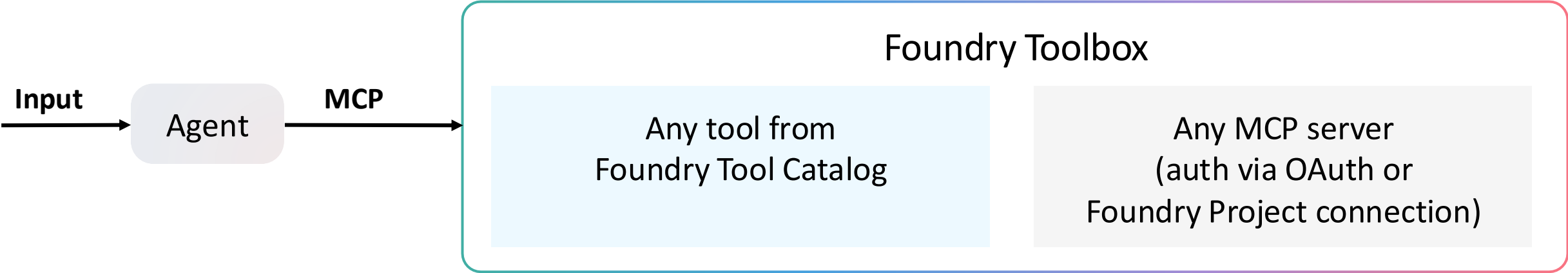


Full example: [src/agent-foundryiq-mcp/main.py](https://github.com/microsoft/agent-foundryiq-mcp/blob/main/main.py)

Foundry Toolbox



Create a toolbox on Foundry composed of Foundry Tools and MCP servers, for use by any agent.



Creating a Foundry Toolbox from code

```
tools = [  
    WebSearchToolboxTool(name="web_search", description="Search the web for current information."),  
    CodeInterpreterToolboxTool(name="code_interpreter", description="Run Python in a sandbox."),  
    MCPToolboxTool(server_label="knowledge-base",  
        server_url=knowledge_base_mcp_url,  
        server_description="Retrieve grounded company and HR information.",  
        project_connection_id=search_connection_name,  
        allowed_tools=["knowledge_base_retrieve"],  
        require_approval="never")]  
  
project = AIPProjectClient(endpoint=endpoint, credential=credential)  
  
version = project.toolboxes.create_version(  
    name=toolbox_name, tools=tools, description=toolbox_description)  
  
project.toolboxes.update(name=toolbox_name, default_version=version.version)
```

 Full code: [infra/create-toolbox-foundryiq.py](#)

 Toolbox <> KB connection made in: [infra/core/ai/ai-project.bicep](#)

Using Foundry Toolbox inside agent

Microsoft Agent Framework has built-in support for the Foundry Toolbox MCP server:

```
toolbox = FoundryToolbox(  
    url=f"{PROJECT_ENDPOINT}/toolboxes/{TOOLBOX_NAME}/mcp?api-version=v1",  
    credential=credential,  
    load_prompts=False)  
  
agent = Agent(  
    client=client,  
    name="InternalHRToolboxHelper",  
    instructions="""You are an internal HR helper focused on employee benefits and company info.  
Use the knowledge base tool to answer company questions and ground answers in provided context.  
Use web search only when the knowledge base does not contain the needed current information.  
Use code interpreter when calculations or structured data analysis would improve the answer."""  
    tools=[toolbox])
```

 Full example: [src/agent-toolbox-foundryiq/main.py](https://github.com/microsoft/agent-toolbox-foundryiq/blob/main/main.py)

Using Foundry IQ inside Toolbox

DEMO

What benefits are available, and when do I need to enroll?

knowledge-base__knowledge_base_retrieve

Input

{"queries":["What benefits are available, and when do I need to enroll?"]}

Output

[{"ref_id":0,"content":"# Contoso\r\n\r\nDisclaimer: This content is generated by AI and may not accurately represent factual information about any real entity. Use this information with caution and verify details from reliable sources.*\r\n\r\n## History\r\n\r\nContoso, a pioneering force in the industry, was founded in 1985 by visionary entrepreneurs with a passion for innovation. Over the years, the company has played a pivotal role in shaping the landscape of consumer electronics.\r\n\r\nYear | Milestone |-----|-----|\r\n\r\n1985 | Company founded with a focus on cutting-edge technology |\r\n\r\n1990 | Launched the first-ever handheld personal computer |\r\n\r\n2000 | Introduced groundbreaking advancements in AI and robotics |\r\n\r\n2015 | Expansion into sustainable and eco-friendly product lines |\r\n\r\n## Company Overview\r\n\r\nAt Contoso, we take pride in fostering a dynamic and inclusive workplace. Our dedicated team of experts collaborates to create innovative

> get_enrollment_deadline_info

> get_current_date

Here's what I found from the available HR benefits materials:

Available benefits

- Health insurance plans through Northwind Health
 - Northwind Health Plus
 - Medical, vision, and dental
 - Prescription drug coverage
 - Mental health and substance abuse coverage

Run local agent:

```
azd ai agent run agent-toolbox-foundryiq
```

Ask question with playground or CLI:

```
azd ai agent invoke agent-toolbox-foundryiq  
--local "What benefits are available, and  
when do I need to enroll?"
```



Full example: [src/agent-toolbox-foundryiq/main.py](https://github.com/microsoft/agent-toolbox-foundryiq/blob/main/main.py)

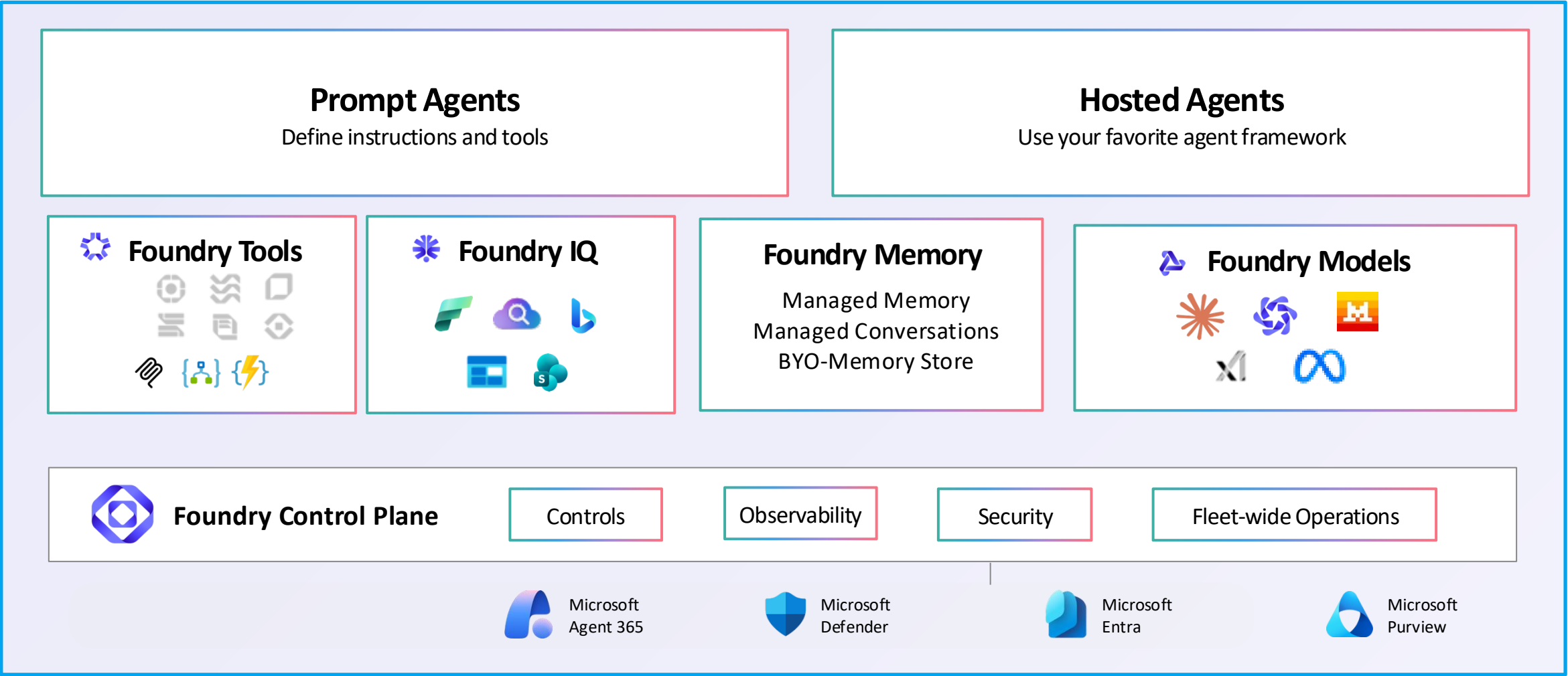
Agent deployment



Foundry Agent Service



Host your agents with production-grade security, reliability, and governance.



Prompt Agents

Define instructions and tools

Hosted Agents

Use your favorite agent framework



Foundry Tools



Foundry IQ



Foundry Memory

Managed Memory
Managed Conversations
BYO-Memory Store



Foundry Models



Foundry Control Plane

Controls

Observability

Security

Fleet-wide Operations



Microsoft
Agent 365



Microsoft
Defender



Microsoft
Entra



Microsoft
Purview

Foundry Hosted Agents

Serve any agent using the Responses API or more generic invocations API.

Foundry Agent Service

Identity · Endpoint · State Scaling · Observability



Azure Container Apps Sandbox



Responses API Adapter

`/agents/{name}/endpoint/protocols/openai/responses`

Your code



Foundry Hosted Agents with MAF

Microsoft Agent Framework provides a built-in Adapter – just wrap your agent and run.

```
from agent_framework import Agent
from agent_framework.foundry import FoundryChatClient
from agent_framework_foundry_hosting import ResponsesHostServer

client = FoundryChatClient(
    project_endpoint=PROJECT_ENDPOINT, model=MODEL_DEPLOYMENT_NAME, credential=credential)

agent = Agent(
    client=client,
    name="InternalHRHelper",
    instructions="You are an internal HR helper.",
    tools=[get_enrollment_deadline_info, get_current_date, toolbox_mcp_tool],
    default_options={"store": False})

server = ResponsesHostServer(agent)
server.run()
```



Full example: <src/agent-toolbox-foundryiq/main.py>

Using hosted agent on Foundry

DEMO

The screenshot shows the Microsoft Foundry interface for an AI agent named 'agent-toolbox-foundryiq'. The left sidebar contains navigation options: Agents, Deployments, Fine-tune, Services, Tools, Knowledge, Memory, Data, Evaluations, and Guardrails. The main area is divided into a 'Chat' tab and a 'Log stream' tab. In the 'Chat' tab, a user has asked 'What benefits are available, and when do I need to enroll?'. The agent's response provides a concise overview of benefits and lists 'Benefits available' for 'Northwind Health Plus', including medical, dental, vision, prescription drug coverage, preventive care, mental health, emergency services, and access to in-network providers. The 'Log stream' tab shows a series of log entries with timestamps and session IDs, indicating the agent's internal processing and connection to the container.

Deploy agent

```
azd deploy agent-toolbox-foundryiq
```

Ask question with playground or CLI:

```
azd ai agent invoke agent-toolbox-foundryiq  
"What benefits are available, and when do I  
need to enroll?"
```

 Full example: [src/agent-toolbox-foundryiq/main.py](https://github.com/microsoft/ai-project-pf-foundryiq/blob/main/main.py)

Use hosted agent in Teams

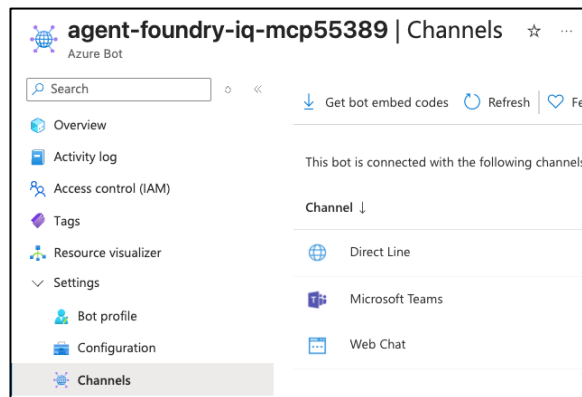
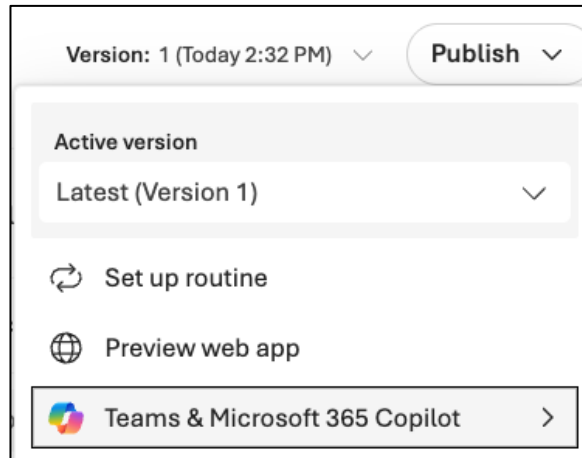
DEMO

Every hosted agent includes the option to publish to Teams.

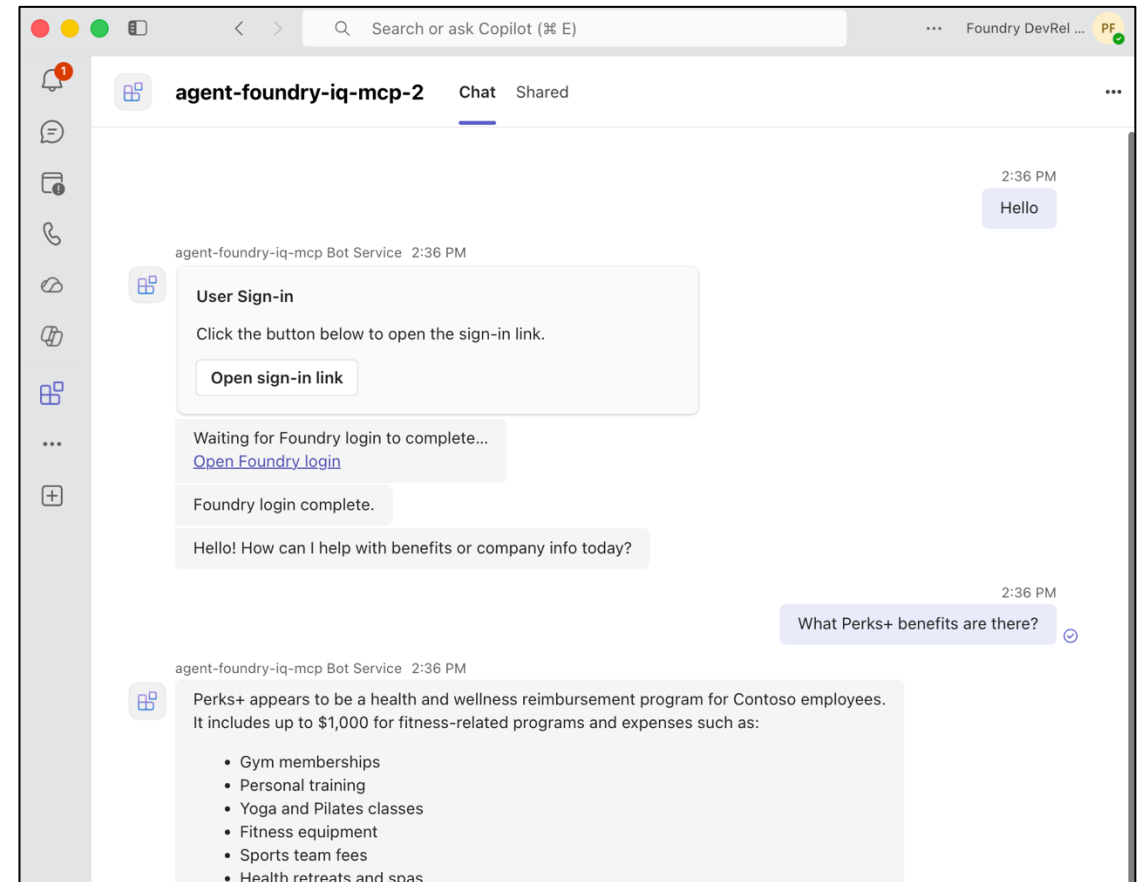
1 Publish from
Foundry UI:



2 Azure Bot
service is
auto-created:



3 Chat with the agent in Teams:



Next steps

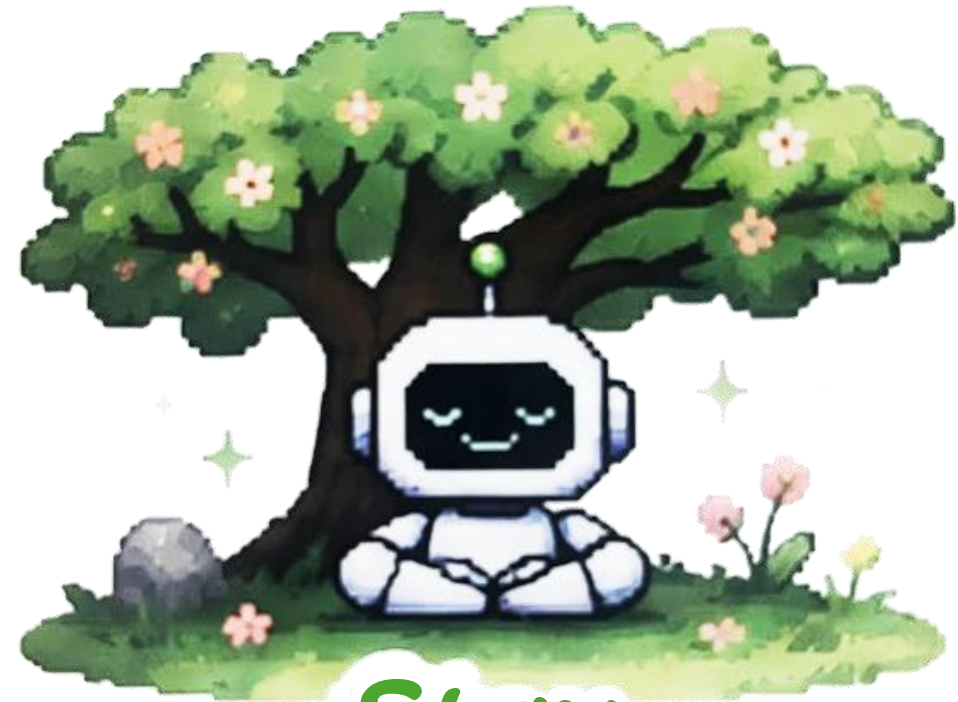
Join office hours after in Discord:
aka.ms/pythonai/oh

Grab all session resources:
aka.ms/iqdeepdive/resources

July 28: Foundry IQ

July 29: Work IQ

July 30: Fabric IQ



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